AIML Lecture 1 MCQ

Foundations and Evolution of AI

1. **[Medium]** Which historical development, though not AI itself, is recognized for paving the way for programmable machines and computational thinking foundational for AI?
   a) The Dartmouth Workshop
   b) The development of FORTRAN
   c) ENIAC (Electronic Numerical Integrator and Computer)
   d) The invention of the transistor
2. **[Medium]** In what year was the Dartmouth Workshop, considered the official birth of the AI field, held?
   a) 1946
   b) 1956
   c) 1966
   d) 1976
3. **[Medium]** Who among the following was NOT explicitly mentioned as an organizer of the 1956 Dartmouth Workshop?
   a) John McCarthy
   b) Marvin Minsky
   c) Claude Shannon
   d) Alan Turing
4. **[Medium]** What was the primary objective of the 1956 Dartmouth Workshop?
   a) To develop the first self-learning algorithm.
   b) To commercialize early AI prototypes.
   c) To find how to make a machine that can simulate every aspect of learning or any other feature of intelligence.
   d) To establish a new programming language for intelligent systems.
5. **[Hard]** The introduction of the term “Artificial Intelligence” formally occurred at which pivotal event?
   a) The development of the first neural network.
   b) The publication of Alan Turing's "Computing Machinery and Intelligence."

c) The creation of ENIAC.
d) The 1956 Dartmouth Workshop.

Traditional Programming vs. AI/ML

1. **[Medium]** In traditional programming, how is the logic for processing data typically handled?
   a) It is inferred from examples by the machine.
   b) It is learned from data using algorithms.
   c) A programmer explicitly writes rules to process data.
   d) It is generated dynamically based on input patterns.
2. **[Medium]** Which of the following is a characteristic of traditional programming (rule-based systems)?
   a) The system learns patterns from data.
   b) The logic is hand-coded, with every condition and action predefined.
   c) The system improves performance without explicit programming for each task.
   d) It relies on labeled examples to classify new inputs.
3. **[Hard]** Consider a program designed to classify a number as even or odd. This is presented in the sources as an example of:
   a) An AI/ML-based system where the model learns from data.
   b) A reinforcement learning problem.
   c) Traditional programming, where the rule is directly embedded in the code by a human.
   d) Deep Learning using a neural network.
4. **[Medium]** How do AI/ML-based systems typically derive their logic?
   a) Through explicit, hand-coded rules provided by a programmer.
   b) By strictly following predefined conditions and actions.
   c) It is inferred from data, not explicitly coded.
   d) By relying solely on human-written rules and output.
5. **[Hard]** What is the fundamental "Key Shift in Programming Paradigm" highlighted when comparing traditional programming with AI/ML-based systems?
   a) The move from compiled languages to interpreted languages.
   b) The shift from single-processor systems to parallel computing.

c) The transition from explicitly coded logic to logic inferred from data using algorithms.
d) The change from input-output systems to interactive user interfaces.

AI vs. ML vs. DL vs. Generative AI - Definitions & Scope

1. **[Medium]** Which term is described as the broad field of computer science focused on building systems that can simulate human intelligence?
   a) Machine Learning
   b) Deep Learning
   c) Generative AI
   d) Artificial Intelligence
2. **[Medium]** Which of the following capabilities is listed as a characteristic of Artificial Intelligence?
   a) Automatically extracting features from raw data.
   b) Creating new content like text and images.
   c) Problem-solving and decision-making.
   d) Learning by interacting with an environment through rewards.
3. **[Medium]** Machine Learning is described as a subset of:
   a) Deep Learning
   b) Generative AI
   c) Artificial Intelligence
   d) Traditional Programming
4. **[Medium]** What is the key idea behind Machine Learning?
   a) Being explicitly programmed for each task.
   b) The system learns patterns and rules from examples rather than being told how to perform a task.
   c) Always using multi-layered neural networks.
   d) Focusing solely on creating new content.
5. **[Medium]** Deep Learning is a subset of:
   a) Artificial Intelligence only
   b) Generative AI only
   c) Machine Learning
   d) Traditional Programming

6. **[Medium]** What is a key characteristic of Deep Learning systems mentioned in the sources?
   a) They only work with small, structured datasets.
   b) They explicitly require human-written rules for every condition.
   c) They use multi-layered neural networks to learn from data.
   d) They primarily rely on labeled data and cannot extract features automatically.
7. **[Hard]** Which concept is positioned as the "umbrella term" under which Machine Learning, Deep Learning, and Generative AI fall?
   a) Data Science
   b) Computer Vision
   c) Artificial Intelligence
   d) Natural Language Processing
8. **[Medium]** Which of these is an example of a capability of Artificial Intelligence, as per the sources?
   a) Training a model with a labeled dataset for classification.
   b) Using multi-layered neural networks to extract features.
   c) Voice assistants like Siri or Alexa.
   d) Generating new images using diffusion models.
9. **[Hard]** While both Machine Learning and Deep Learning enable computers to learn from data, what distinguishes Deep Learning in terms of its core approach?
   a) ML focuses on supervised learning, while DL focuses on unsupervised learning.
   b) ML requires explicit feature engineering, while DL automatically extracts features using multi-layered neural networks.
   c) DL is a broader field than ML.
   d) ML prioritizes creating new content, while DL focuses on prediction.
10. **[Medium]** What is Generative AI specialized in?
    a) Recognizing objects and classifying data.
    b) Optimizing resource allocation in complex systems.
    c) Creating new content such as text, images, music, and code.
    d) Analyzing historical data for predictive analytics.

Types of Machine Learning

1. **[Medium]** What type of Machine Learning algorithm learns from a labeled dataset, where each training example is paired with the correct output?
   a) Unsupervised Learning
   b) Reinforcement Learning
   c) Supervised Learning
   d) Deep Learning
2. **[Medium]** Which of the following is a common task for Supervised Learning?
   a) Clustering similar data points.
   b) Discovering hidden patterns in unlabeled data.
   c) Regression, to predict numerical values.
   d) Learning by trial and error through rewards.
3. **[Medium]** Email spam detection (classifying emails as "spam" or "ham") is given as an example of which type of Machine Learning?
   a) Unsupervised Learning
   b) Reinforcement Learning
   c) Supervised Learning
   d) Generative AI
4. **[Medium]** If a Machine Learning algorithm works on unlabeled data and tries to find hidden patterns or structures without explicit instructions, what type of learning is it?
   a) Supervised Learning
   b) Unsupervised Learning
   c) Reinforcement Learning
   d) Semi-supervised Learning
5. **[Medium]** Grouping customers based on buying behavior is an example of which Unsupervised Learning technique?
   a) Regression
   b) Clustering
   c) Classification
   d) Dimensionality Reduction
6. **[Medium]** What is the goal of Unsupervised Learning?
   a) To predict outputs for new, unseen data.
   b) To take actions that maximize cumulative rewards over time.

c) To discover groupings, associations, or patterns in data.
d) To learn from labeled datasets with correct outputs.

7. **[Medium]** In Machine Learning, the analogy "Like learning with an answer key" is used to describe which type of learning?
   a) Unsupervised Learning
   b) Reinforcement Learning
   c) Supervised Learning
   d) Deep Learning
8. **[Medium]** The analogy "Like exploring a puzzle with no hints" is used to describe which type of Machine Learning?
   a) Supervised Learning
   b) Unsupervised Learning
   c) Reinforcement Learning
   d) Generative AI
9. **[Medium]** Which type of Machine Learning model learns by interacting with an environment and receiving rewards or penalties?
   a) Supervised Learning
   b) Unsupervised Learning
   c) Reinforcement Learning
   d) Deep Learning
10. **[Medium]** In Reinforcement Learning, what is the 'Agent'?
    a) The feedback from the environment.
    b) The world it interacts with.
    c) The learner or decision-maker.
    d) The strategy used to determine the next action.
11. **[Medium]** Game AI, such as AlphaGo or Dota 2 bots, are given as examples of which type of Machine Learning?
    a) Supervised Learning
    b) Unsupervised Learning
    c) Reinforcement Learning
    d) Deep Learning
12. **[Medium]** What is the goal of Reinforcement Learning?
    a) To predict categories or numerical values based on labeled data.

b) To find hidden structures in unlabeled data.
c) To take actions that maximize cumulative rewards over time.
d) To generate new, creative content.

13. **[Hard]** Consider a scenario where a company wants to segment its customer base for targeted marketing campaigns without pre-defined categories. Which Machine Learning type would be most appropriate for this task, and what technique would it likely employ?
    a) Supervised Learning, using Classification.
    b) Reinforcement Learning, optimizing rewards for customer engagement.
    c) Unsupervised Learning, using Clustering.
    d) Deep Learning, leveraging RNNs.
14. **[Hard]** A model designed to recommend products to users based on their similarity to other users' preferences, without explicit labels for "good" or "bad" recommendations, would primarily fall under which ML type?
    a) Supervised Learning (Regression)
    b) Reinforcement Learning (Agent-Environment interaction)
    c) Unsupervised Learning (Product recommendation using similarity)
    d) Deep Learning (CNNs for image similarity)
15. **[Hard]** The "vanishing gradient problem" is a challenge often associated with traditional RNNs. Which variant of RNNs was developed to specifically address this issue?
    a) GRU
    b) LSTM
    c) Transformers
    d) CNN

Deep Learning Models

1. **[Medium]** Which Deep Learning architecture is primarily used for image processing and computer vision tasks?
    a) Recurrent Neural Networks (RNNs)
    b) Transformers
    c) Convolutional Neural Networks (CNNs)
    d) Generative Adversarial Networks (GANs)

2. **[Medium]** How do Convolutional Neural Networks (CNNs) primarily work to detect patterns in images?
   a) By maintaining a memory of previous inputs via loops.
   b) By using a self-attention mechanism to process tokens in parallel.
   c) By applying filters (kernels) over input images to detect patterns.
   d) By competing between a generator and a discriminator.
3. **[Medium]** Handwritten digit recognition (MNIST) and face detection are provided as examples of applications for which Deep Learning model?
   a) RNNs
   b) Transformers
   c) CNNs
   d) GANs
4. **[Medium]** Which Deep Learning architecture is designed specifically for sequential data such as text, speech, and time series?
   a) Convolutional Neural Networks (CNNs)
   b) Recurrent Neural Networks (RNNs)
   c) Transformers
   d) Diffusion Models
5. **[Medium]** How do Recurrent Neural Networks (RNNs) maintain context from previous inputs?
   a) Through pooling layers.
   b) By applying convolutional filters.
   c) Via loops in their architecture that maintain a memory.
   d) By processing all tokens in parallel.
6. **[Medium]** Language modeling and sentence generation, as well as speech-to-text transcription, are examples of applications for which Deep Learning model?
   a) CNNs
   b) RNNs
   c) Transformers
   d) GANs
7. **[Medium]** What is the key innovation of Transformers that allows the model to focus on different parts of the input simultaneously?
   a) Pooling layers

b) Recurrent loops
c) Generative adversarial learning
d) Self-attention mechanism

8. **[Medium]** Which Deep Learning architecture is foundational for modern large language models (LLMs) like GPT and BERT?
a) CNNs
b) RNNs
c) Transformers
d) GANs

9. **[Medium]** What is a significant advantage of Transformers over RNNs, as stated in the sources?
a) Transformers are simpler in architecture.
b) RNNs are more accurate for language tasks.
c) Transformers process all tokens in parallel, leading to faster and more scalable operations.
d) RNNs inherently handle the vanishing gradient problem better.

10. **[Hard]** A medical image analysis system that identifies cancerous cells in X-ray scans would most likely utilize which type of Deep Learning architecture?
a) Recurrent Neural Network (RNN)
b) Transformer
c) Convolutional Neural Network (CNN)
d) Generative Adversarial Network (GAN)

11. **[Hard]** A stock price prediction model that considers historical market data as a time series would be best suited for which deep learning architecture among the following, given its ability to handle sequential data?
a) CNNs
b) RNNs
c) Transformers
d) Diffusion Models

12. **[Hard]** In the context of NLP tasks like language translation and document summarization, Transformers are considered state-of-the-art. This is primarily due to their ability to:
a) Reduce dimensionality through pooling layers.
b) Process input tokens in parallel using self-attention, capturing long-range

dependencies efficiently.
c) Maintain an internal memory of previous inputs via recurrent loops.
d) Generate photorealistic images by competing networks.

Generative AI

1. **[Medium]** What kind of content can Generative AI create?
   a) Only text and code.
   b) Only images and music.
   c) Text, images, music, and code.
   d) Only existing data classifications.
2. **[Medium]** Which of these tools is specifically mentioned for image generation using Generative AI?
   a) ChatGPT
   b) GitHub Copilot
   c) DALL·E
   d) AlphaGo
3. **[Medium]** Generative Adversarial Networks (GANs) consist of two competing neural networks. What are they called?
   a) Encoder and Decoder
   b) Classifier and Regressor
   c) Generator and Discriminator
   d) Input and Output
4. **[Medium]** Which Generative AI technology works by adding noise to data and then learning to reverse that noise to produce high-quality images?
   a) GANs
   b) Transformer Models
   c) Diffusion Models
   d) Recurrent Neural Networks
5. **[Medium]** Which Generative AI technology is exemplified by models like GPT-4 and Gemini?
   a) GANs
   b) Diffusion Models
   c) Transformer Models
   d) Convolutional Neural Networks

6. **[Medium]** Deepfake video creation and AI-generated clothing designs are given as examples of applications for which generative model?
   a) Diffusion Models
   b) Transformers
   c) GANs
   d) RNNs
7. **[Medium]** Tools like DALL·E 2, Midjourney, and Stable Diffusion are powered by which Generative AI technology?
   a) GANs
   b) Diffusion Models
   c) Transformer Models
   d) Recurrent Neural Networks
8. **[Hard]** Generative AI is capable of producing new content. In the context of a "Key Shift in Programming Paradigm," how does this capability fundamentally differ from traditional programming or even typical discriminative AI/ML tasks?
   a) It still relies on explicit rules to create content, similar to traditional programming.
   b) It primarily focuses on classifying existing data, which is a core ML task.
   c) It actively produces novel output (text, images, etc.) rather than merely processing or classifying existing inputs or inferring rules from them.
   d) It is limited to understanding and processing natural language.

Prompt Engineering

1. **[Medium]** What is "Prompt Engineering"?
   a) The process of designing AI models.
   b) The art of debugging AI code.
   c) The skill of writing clear and effective prompts to get useful results from AI tools.
   d) The method of training AI models with large datasets.
2. **[Medium]** Why is Prompt Engineering considered important?
   a) It directly influences the speed of AI model training.
   b) It determines the hardware requirements for AI tools.
   c) The quality of the AI output depends on the quality of the prompt.
   d) It allows for the selection of the best AI architecture for a task.

3. **[Medium]** Which of the following is a recommended tip for writing great prompts?
   a) Keep prompts as short as possible, avoiding details.
   b) Be clear and specific, and mention the desired style, tone, or format.
   c) Avoid adding examples, as they can confuse the AI.
   d) Focus only on the output, not the input.
4. **[Hard]** A user wants to generate an image of a futuristic city at sunset with a cyberpunk aesthetic using DALL·E. Which prompt engineering tip would be most crucial for achieving the desired visual style?
   a) Making the prompt as short as possible.
   b) Focusing solely on the object (futuristic city).
   c) Mentioning the desired style and tone (cyberpunk aesthetic, sunset).
   d) Providing an example of an unrelated image.
5. **[Hard]** If an AI tool like ChatGPT is generating generic or unhelpful responses, what is the most likely reason related to prompt engineering, as per the sources?
   a) The AI model itself is inherently flawed.
   b) The user's internet connection is too slow.
   c) The prompt was not clear, specific, or lacked necessary context/style instructions.
   d) The AI tool has run out of its creative capacity.